TOWARDS INTELLIGENT MUSIC EDUCATION: SCORE-INFORMED TRANSCRIPTION AND PERFORMANCE ASSESSMENT

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ABSTRACT

We present work in progress on a score-informed automatic music transcription and performance assessment system designed to support automated musical education. The system combines transcription models with score alignment and error detection tools to provide meaningful feedback to both students and teachers. It comprises an automatic guitar transcription model trained using a combination of public and internal data, along with pre-trained transcription models for piano and violin. Following transcription, the system evaluates performances by aligning the transcribed results with reference music scores to identify various errors, such as pitch, rhythm, tempo, structural, and intonation. This work demonstrates how intelligent systems can enhance the music learning experience by making it more interactive, personalized, and data-driven.

1. INTRODUCTION

In recent years, advances in artificial intelligence have significantly transformed the field of music education, introducing innovative tools to improve the learning experience for both teachers and students. AI-assisted music education systems blend traditional teaching approaches with intelligent technologies capable of analyzing, evaluating, and responding to musical performances in real time. These systems are composed of several key components, including automatic transcription, performance assessment, and personalized feedback. By integrating these elements, AI-driven tools can make music practice more interactive, engaging, and adaptive. As a result, teachers are empowered to provide more individualized instruction tailored to each student's progress and specific needs, while students benefit from immediate, data-driven feedback during their practice sessions, helping them to self-correct and develop more efficiently.

In this context, our goal is to develop an AI-assisted system that provides students with automated and meaningful feedback on their instrumental performances. Focusing on three widely studied instruments including piano, violin, and guitar, our system takes an audio recording of a student's performance, transcribes it into symbolic form, aligns it with the original music score, and then detects and reports mistakes such as pitch, timing, or rhythmic errors.

As presented in Figure 1, our approach consists of three key components: **(1)** automatically transcribing guitar, piano, and violin recordings into MIDI format; **(2)** automatically aligning the ground truth score with the MIDI transcription of the student's performance; and **(3)** detecting performance errors mismatches identified in the aligned sequences from step 2.

2. METHODOLOGY

2.1 Transcription

Our transcription work initially focuses on guitar. We leverage a private internal dataset of 2.8 hours of fully annotated guitar recordings produced by Algorivm [1] for our training data. Our final dataset is a combination of this internal data and GuitarSet [1] for a total of 5.8 hours of training data. Because human performances tend to have many small timing deviations from notes in a score, we first create a fine-alignment of MIDI notes to note onsets in order to better match the note timings in our ground truth data annotations to the performance. This was done using an initial stage of dynamic time warping, followed by a fine-alignment stage which further aligns the notes using a pre-trained transcription model [2]. Our guitar transcription model is based on the high-resolution piano transcription model of Kong et al. [3]. It is first trained on the MAESTRO dataset [4] using various data augmentations as seen in [5] and then fine-tuned on the mixed dataset described above. When evaluated on the test split of GuitarSet and our internal dataset, our model achieved precision, recall and f1 scores of 0.8895, 0.9067 and 0.8973 respectively at the frame level and 0.8903, 0.8926 and 0.8900 at the note level. Our final system also incorporates the high-resolution transcription model proposed by Tamer et al. [6] for violin, and the model introduced by Kong et al. [3] for piano.

[1] https://algorivm.com/

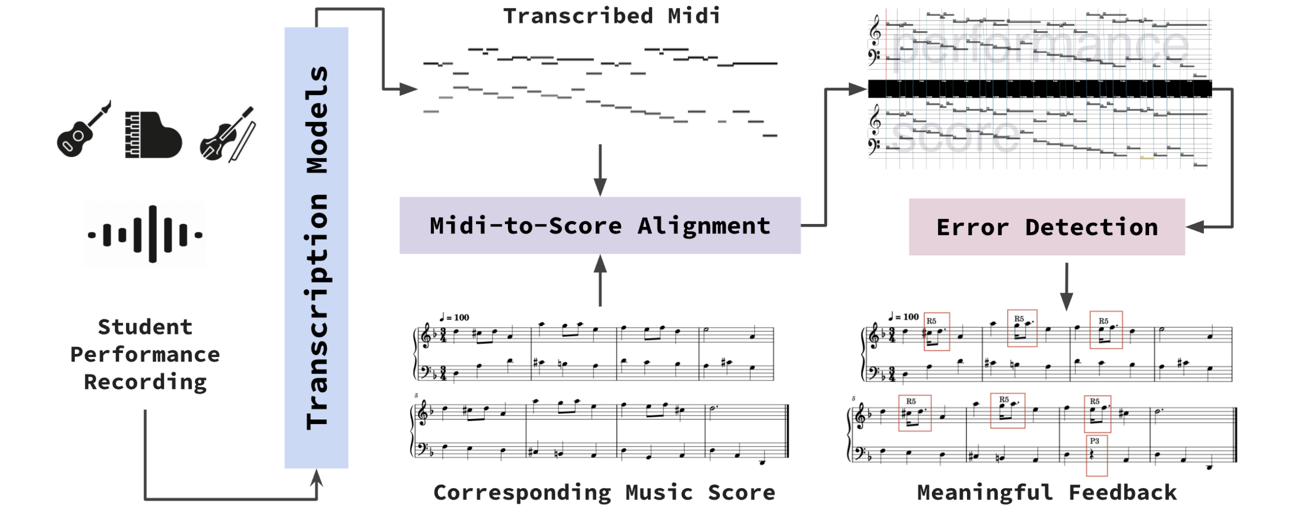

Figure 1. Overview of the proposed system. The top right panel shows the alignment between a transcribed piano performance (in MIDI format) and the original score using Parangonada [7]. The excerpt includes the first eight bars of Henry Purcell's *Air in D Minor*. The feedback panel annotates performance errors detected by our algorithm. **R5** indicates a sequence played with correct timing but incorrect rhythmic values, while **P3** marks a missing note that does not affect surrounding musical content.

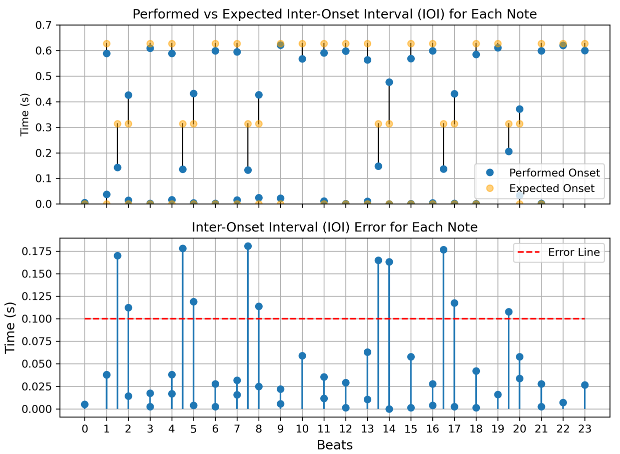

Figure 2. Detection of rhythmic and tempo deviations based on inter-onset intervals (IOIs) for the performance presented in Figure 1. The top plot represents a comparison between performed and expected IOI per note, with vertical lines indicating deviations, while the bottom plot shows absolute IOI errors, with the red dashed line representing the error threshold.

2.2 Performance Assessment

To align the transcribed MIDI files with the original music scores, we use Parangonada [7], a tool which leverages multiple dynamic time warping algorithms as illustrated in Figure 1. This approach provides note-wise alignments between the score and performance, which serve as the basis for subsequent analysis. Our error detection in musical performances focuses on five main categories: pitch, rhythm, tempo, structural, and intonation errors, each further divided into specific subcategories. Since intonation errors are not relevant to piano performances, we consider them only in the context of guitar and violin.

Pitch errors, such as missing and extra notes, are directly extracted from the alignment results. Rhythmic and tempo-related errors are identified by comparing expected and observed inter-onset intervals (IOIs), where IOI refers to the time between successive note onsets. The global tempo is estimated by averaging the ratio between each note's duration in seconds and its duration in beats. Based on this tempo estimation, we compute the expected onset time of each note and derived its expected IOI. As illustrated in Figure 2, a rhythmic error is flagged for nth-note when the difference between the expected and observed IOIs exceeded a threshold δ:

$$|\mathrm{IOI}_n^{\text{expected}} - \mathrm{IOI}_n^{\text{observed}}| > \delta \tag{1}$$

In future work, we will focus on extending the system to detect errors such as structural deviations and intonation errors, and on classifying detected errors in a more formative manner.

3. CONCLUSION

We have presented progress on a score-informed transcription and performance assessment for the purposes of improving musical education. By integrating transcription, alignment, and error detection into a unified multi-instrument framework, our system provides students and educators with personalized feedback. Future work will focus on improving transcription accuracy using synthesized data and developing a more sophisticated assessment pipeline. We hope that this demo showcases the potential of such tools to make music learning more effective and inspires further work in intelligent music education systems.

4. ACKNOWLEDGMENTS

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